**THE WISE ONES**

**Client problem + additional focus**: Where does the trash come from (trends)

**Tentative waste-audit location**: Art A & B (Library)

**Tentative type of waste audit**: Landfill

**Deliverables:**
**Relational Database Prototype**
**Spreadsheet Prototype**
**Storage System Evaluation**
**Financial Impact Metric**
**Environmental Impact Metric**
**Data-Collection Recommendation**

- what variables should be collected
    - how to define and record
- how much data to collect
- where/when data should be collected

**Additional client problem**

**Final Report**

Estimated Sample Size: 64 samples in both groups (assuming 2 sample t-test)

# Team Roles:

## **Project Manager**: Jimmy Wee Sit

**Job Duties**:

- Assist with duties as needed
- Communication
- Schedule meetings
- Outline of project
- Second line of Quality assurance

**Deliverables assigned to this person:**

- Make sure project is understood and comprehensible
- Turn in assignments
- Due dates

## **Technical Lead:** Strawberry

**Job Duties:**

- Lead design of relational database schema and spreadsheet
- Coordinate construction of relational schema and spreadsheet

**Deliverables assigned to this person:**
Database Prototypes

### **Quality Assurance:** Roxy

**Job Duties:** Verifying datasets, making sure models are accurate, ensuring there are no errors, data validation, model training and validation
Deliverables assigned to this person: data collection recommendation, additional client problem

### **Domain Expert:** Chase

**Job Duties**: Background research into client problem and university standing/ goals on waste. Identify information needed to develop database, spreadsheet, impact metrics. Basically research/gather any information the team needs to successfully solve the clients problem and create a consolidated research record. Schedule and meet with necessary parties, record notes.
Deliverables assigned to this person: Financial and Environmental metrics

# Shared Work:

**What client problem or need will your team investigate in addition to the required deliverables?**
Special waste categorization.

**Why does this problem matter to the client or to the goal of reducing campus waste?**
If the client knows what specific waste is there, they can identify the source of it.

**What waste-audit data, campus information, or prototype feature could help your team investigate this problem?**
We can make a column for most common waste found in the audit.

**What does your team expect to address during the next three weeks?**
Week 1: design database and Storage
Week 2: Actual Data collection
Week 3: refinements and reports

**What related issues are outside the scope of this project?**
Social impacts

Audit Location:
**Why this location?** Art supplies are specialized items that could be identified easier.
**What is included within the physical boundary of this location**
**Who controls or manages the location?** Morgan King
**What access or permission will the team need?** Access to the dumpsters

Audit Type:
**What waste stream, source, or activity will your team examine?** Dumpsters and trashcans
**What will the team observe, sort, count, weigh, or otherwise measure?** Trash

**What will be the unit of observation? Examples include a bag, bin, collection, station, building day, event, or individual item.**
Bag or bin
**What waste categories will need to be recorded?**
Landfill, compost, recyclable, hazardous waste
**What timing, frequency, or number of audit occasions are you currently considering?**
2-3x same location

Relational Database Prototype:
Work required:
Primary owner: Strawberry
Supporting team members: Chase, Roxy

Spreadsheet Prototype:
Work required: wait till Database designed
Primary owner: Jimmy
Supporting team members: Roxy, Strawberry

Storage-System Evaluation:
Work required:
Primary owner: Roxy
Supporting team members: Chase, Strawberry, Jimmy

Financial Impact metric: How much does it cost to move waste
Work required:
Primary owner: Chase
Supporting team members: Jimmy

Environmental Impact metric:

Work required:
Primary owner: Chase
Supporting team members: Jimmy

Data Collection Recommendation (methodology):
Work required:
Primary owner: Roxylynn
Supporting team members: Chase

Additional Client Problem:
Work required:
Primary owner: Roxylynn
Supporting team members: Strawberry

Final report:
Work required:
Primary owner: Jimmy
Supporting team members: Chase, Roxy, Strawberry

# Three Week Timeline:

Week 1: Scope and Design
- confirm audit location and type
- identify special focus
- identify the variables required
- draft relational database schema/spreadsheet
- select preliminary financial and environmental impact metrics
- identify missing data, permissions, costs, or conversion factors
- Background research on client problem

Week 2: Builds, Tests, Collect data
- Build database/spreadsheet prototypes
- enter/generate enough data to test systems
- develop impact calculations
- gather evidence related to the special focus
- begin documenting design decisions and limitations

Week 3: Evaluate and Report
- Evaluate the 2 storage systems
- Select and justify evaluation criteria

- complete impact examples
- finalize the data-collection methodology
- complete the analysis related to the special focus
- integrate work into report
- check integrations
- review everything

**Dependencies:**
What tasks cannot begin until another task, decision, permission, or data source is complete?

Background Research → Variables/Metrics → Database Design → Data Collection → Reports → Final Report